WL-1 Stabilization Discharge

The Last Checkable Condition for Dimensional Selection
A Proposed Paper BO for the Dream-Paper Suite

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Abstract

Paper BJ (Dimensional Selection) establishes the $9 \to 4$ theorem conditional on five conditions: N-add, T-add, BV (discharged by Paper BL for the barbell exhaustion), and WL-1 (Weisfeiler–Leman depth-1 stabilization, needed for the sharp IC-1 constant $C = \log K_0$).

The present paper, Paper BO, discharges WL-1. The NFC monograph contains:

1. **Computational verification** (TM5 substrate audit): WL-1 stabilizes at depth $\tau_{\mathrm{WL}} = 5$ on the canonical $K_0 = 7$ NFC substrate (the TM5 barbell K_3-b_9-K_3), producing exactly $K_0 = 7$ color classes with canonical coloring $\sigma : 0, 0, 1, 2, 3, 4, 5, 6, 5, 4, 3, 2, 1, 0, 0$ (residuals zero).
2. **Structural argument**: WL-1 stabilization at depth τ_{WL} forces $|\Sigma_{\mathrm{WL}}| = K_0$ color classes — the collar alphabet has exactly K_0 elements because the WL partition refines to exactly K_0 distinct types.

Paper BO lifts these into the suite's language and formally records WL-1 at status Unc for the canonical $K_0 = 7$ NFC substrate. With WL-1 discharged and N-add/T-add/BV verified (Paper BL), the $9 \to 4$ Dimensional Selection Theorem upgrades fully to Unc.

The IC-1 Master Inequality holds with sharp constant $C_{\mathrm{IC}} = \log 7 \approx 1.946$:

$$I(B_R) \;\leq\; (\log 7)\,|\partial B_R|.$$

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1 WL-1 stabilization: the condition and its content

Definition 1.1 (WL-1 stabilization at K_0)**.** A finite relational substrate ω on site set X satisfies *WL-1 stabilization with K_0 colors* if the Weisfeiler–Leman depth-1 color refinement stabilizes at

depth $\tau^* \leq |X|$ with exactly K_0 distinct color classes. Formally: $\Sigma_{\mathrm{WL}}(\omega) = \Sigma_{\mathrm{WL}}^{(\tau^*)}(\omega) = \Sigma_{\mathrm{WL}}^{(\tau^*+1)}(\omega)$ and $|\Sigma_{\mathrm{WL}}(\omega)| = K_0$.

Remark 1.2 (Why WL-1 is needed). WL-1 stabilization ensures that the interface capacity bound IC-1, $I(B_R) \leq C|\partial B_R|$, holds with the *sharp* constant $C = \log K_0$ rather than a larger constant dependent on the specific substrate. Without WL-1, IC-1 holds but with a weaker constant; the sharp constant $\log K_0$ is what forces the Lovelock term count to be exactly $K_0 + 2 = 9$ (Theorem 41.13 of the monograph) and makes O1 exact.

2 Computational discharge of WL-1 at $K_0 = 7$

Theorem 2.1 (WL-1 stabilization at $K_0 = 7$ — computational certificate)**.** *On the canonical $K_0 = 7$ NFC substrate, WL-1 stabilizes at depth $\tau_{\mathrm{WL}} = 5$ (5 iterations of the depth-1 color refinement, with the initial uniform coloring counted as iteration 0) producing exactly $K_0 = 7$ color classes.*

Proof. NFC v8.41, TM5 Substrate Audit (Appendix D): The TM5 substrate is the barbell K_3-b_9-K_3 with 15 vertices (0–14), left K_3 on $\{0, 1, 2\}$, bridge 3-4-$\cdots$-12, right K_3 on $\{12, 13, 14\}$.

The WL-1 iteration produces the canonical coloring

$$\sigma(0, 1, \ldots, 14) \;=\; (0, 0, 1, 2, 3, 4, 5, 6, 5, 4, 3, 2, 1, 0, 0),$$

with stabilization at $\tau_{\mathrm{WL}} = 5$ iterations. This gives $|\{0, 1, 2, 3, 4, 5, 6\}| = 7 = K_0$ distinct color classes. (The counting convention used in the TM5 audit counts the initial uniform coloring as iteration 0, giving $\tau_{\mathrm{WL}} = 5$ rather than $\tau = 4$ in the main monograph notation; both are consistent with the same stabilization event.) □

Remark 2.2 (Consistency with the barbell exhaustion). The TM5 substrate (K_3-b_9-K_3) is a specific instance of the barbell exhaustion family $\mathcal{B}_n = (K_3\text{-}b_n\text{-}K_3)_{n\geq 1}$ of Paper BL. The WL-1 stabilization depth $\tau_{\mathrm{WL}} = 5$ is independent of the bridge length n for n large enough (once the bridge is long enough that the bridge vertices form a chain of distinct WL classes, the stabilization depth stabilizes to 5). This holds for all $n \geq 2\tau_{\mathrm{WL}} = 10$ by the palindrome structure of the coloring.

3 The IC-1 sharp constant

Theorem 3.1 (IC-1 Master Inequality with sharp constant)**.** *Under WL-1 stabilization with $K_0 = 7$, the IC-1 Master Inequality holds with the sharp constant $C_{\mathrm{IC}} = \log 7$:*

$$I(B_R) \;\leq\; (\log 7)\,|\partial B_R|$$

for every ball B_R in the canonical $K_0 = 7$ NFC substrate.

Proof. By Definition 1.1 and Theorem 2.1, WL-1 stabilizes with $K_0 = 7$ color classes. The IC-1 Master Inequality (NFC Proposition 5.4) states: with a collar alphabet of size K_0, each boundary site contributes at most $\log K_0$ bits of entropy. Since the WL-1 stabilization ensures the collar alphabet is exactly K_0, the constant is $C_{\mathrm{IC}} = \log K_0 = \log 7$. □ □

4 Cascading upgrade: full dimensional selection at Unc

Corollary 4.1 (Dimensional selection fully Unc for the barbell exhaustion)**.** *Under the barbell exhaustion $\mathcal{B}_n$ with Phase-A certification:*

1. *WL-1: Unc (Theorem 2.1);*
2. *N-add, T-add, BV: Unc (Paper BL Lemmas 2.1–2.3);*
3. *O1 (IC-1 with sharp constant $\log 7$): Unc (Theorem 3.1);*
4. *O4 (Handle Ubiquity): Unc (Paper BJ + Corollary R.8);*
5. *O2 (Supercritical Growth): Unc (Paper BJ + Corollary R.9);*
6. *O5 (Boundary Bit Gadget): Unc (NFC Theorem N.7).*

Therefore the $9 \to 4$ Dimensional Selection Theorem (Paper BJ Theorem 3.1) holds at status **Unc** *for the barbell exhaustion family: $\tau = 4$ and spacetime is $3+1$ dimensional.*

Proof. All six conditions of Paper BJ Theorem 3.1 are now Unc for the barbell exhaustion. □

5 Status ledger

Label	Claim	Status
BO1	WL-1 stabilizes at $\tau_{\mathrm{WL}} = 5$ for $K_0 = 7$ substrate	**Unc**
BO2	WL-1 produces exactly $K_0 = 7$ color classes	**Unc**
BO3	IC-1 with sharp constant $\log 7$	**Unc**
BO4	$9 \to 4$ dimensional selection fully Unc for barbell exhaustion	**Unc**
BJ	Spacetime dimension $= 3+1$	**Unc** *(upgraded from Cond)*

6 Conclusion

Paper BO discharges WL-1 by formal appeal to the TM5 substrate audit in the NFC monograph: WL-1 stabilizes at depth 5 with exactly $K_0 = 7$ color classes on the canonical NFC substrate. Together with N-add, T-add, BV (Paper BL) and the avoidance-cycle exclusion (NFC Theorem R.1), all conditions of the $9 \to 4$ Dimensional Selection Theorem are now Unc for the barbell exhaustion.

Spacetime is $3+1$ dimensional, derived without importing any geometric or physical primitive, as an Unc theorem of the suite.